

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Examination (EC/ME/CL/CE/IT/EE)

December 2011

ME 102 Fundamentals of Mechanical Engineering

Date: 07.12.2011, Wednesday, Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION I

- Q:1 (a). What is difference between power and energy? Give their SI units. [02]
(b). Write advantages of non-conventional sources of energy over conventional sources of energy. [03]
(c). What is the practical significance of gas constant? Prove that $C_p - C_v = R$ with usual notations. [05]
- Q:2 (a). Derive the equation for adiabatic process for relation between P, V, & T and work done. [05]
(b). 1 kg of air at a pressure of 1 bar and temperature of 27°C is compressed to pressure of 10 bar adiabatically. Calculate work done, heat transfer and change in internal energy. Take $C_p = 1.005 \text{ kJ/kg K}$, $C_v = 0.718 \text{ kJ/kg K}$. [05]

OR

- Q:2 (a). Explain elementary heat engine with sketch. Write down the classification & names of heat engine cycles. [05]
(b). The following data were collected during the test on a single cylinder four stroke, oil engine:
cylinder diameter = 270 mm, stroke length = 380 mm, Mean effective pressure = 6 bar, Engine speed = 250 rpm, Effective braking torque 750 N.m, Fuel used = 10 Kg/hr, C.V. of fuel = 44400 KJ/Kg
Calculate: (i) Brake Power (ii) Indicated Power (iii) Mechanical Efficiency (iv) Indicated Thermal Efficiency. [05]
- Q:3 (a). Show that the air standard efficiency (η) of Otto cycle is function of Compression Ratio only. [05]
(b). A single stage, single acting reciprocating compressor runs at 330 r.p.m. with piston speed 190 m/min. It has intake pressure 1 bar and delivery pressure 15 bar. The compression and expansion follows the law $PV^{1.3} = C$. Indicated power is 28 KW and volumetric efficiency is 95%. Determine bore and stroke. [05]
(c). Name the different power transmission elements. Classify the Mechanical drives. [05]

OR

- Q:3 (a). Derive an equation for work done in case of a single stage single acting reciprocating air compressor neglecting clearance. [05]
(b). What is function of coupling? Name only various types of couplings. Explain Oldham coupling. [05]
(c). (i) Differentiate brake and clutch. [02]
(ii) Advantages of gear drive over belt drive. [03]

P. T. O.

SECTION II

- Q:4 (a). Explain what you understand by term quality of steam. What are the benefits of superheated steam? [02]
- (b). Determine the enthalpy and specific volume of 1 kg steam at 8 bar (i) when the dryness fraction is 0.9 and (ii) when steam is superheated to a temperature of 280° . Take $CPS=2.1$ KJ/kg-K. [03]
- (c). Explain throttling calorimeter with neat sketch. Also mention its limitations. [05]
- Q:5 (a). Draw the neat sketch of Babcock and Wilcox boiler and explain its working. [05]
- (b). What is stoichiometric air requirement? [05]
- A coal sample gave the following analysis by weight:
Carbon 86%, hydrogen 5%, sulphur 0.8%, oxygen 6% and remainder being incombustible. Determine the minimum air required per Kg of coal for chemically correct composition.
- (c). Steam at 8 bar and dryness fraction of 0.9 expands at constant pressure until the temperature is 200° C. Calculate the work input and heat supplied/Kg of steam during process. [05]

OR

- Q:5 (a). State advantages disadvantages of gaseous fuels. Give comparison of LPG & CNG. [05]
- (b). Enlist the different mountings & accessories of boiler. Draw neat sketch of Air preheater & state its function, location and advantages. [05]
- (c). A Boiler evaporates 17000 kg/hr of steam at 12.5 bar and dryness fraction of 0.97 from feed water at 105° C. When coal is fired at the rate of 2000 kg/hr having calorific value 27400 KJ/kg, find (I) Heat supplied per hour. (ii) The thermal efficiency. (iii) The equivalent evaporation. [05]
- Q:6 (a). Explain the differences between heat engine, heat pump & refrigerator with block diagram [05]
- (b). Explain the main difference between [05]
- (i) Positive displacement pump and rotodynamic pump
- (ii) Single acting and Double acting pump

OR

- Q:6 (a). Explain with a neat sketch the working of a Vapour Compression Refrigeration system. [05]
- (b). Classify the centrifugal pump and Explain the vortex type pump with neat sketch. [05]